

Project Overview

TEAM

Taj Tea-m

PROJECT

Barrier Brain

PROBLEM STATEMENT: RPM HIRE - Digital Tool for Temporary Infrastructure (#3)

THE PROBLEM

Councils can't foresee the knock-on effects of roadworks.

Every roadwork is planned on a **traffic guidance scheme (TGS)**, like this one. Roadworks under four weeks skip traffic modelling, so this drawing is all they have.

It can't tell you:

- how many hours of delay it adds across the city
- which streets traffic spills into
- how many pedestrians are put at risk

Congestion already costs Melbourne \$4.6 billion a year. **So how do we go from a drawing to real answers?**



THE SOLUTION

Barrier Brain turns a TGS into an *actionable report on delays, detours, pedestrian safety and public transport* before a single barrier goes out.

It combines **agentic workflows, photogrammetry and traffic simulation modelling** to tell planners:

- How many hours of delay it adds across the city
- Which streets traffic spills into
- How many pedestrians are put at risk

Councils and contractors foresee the knock-on effects and fix them in the plan. That saves money and time, and keeps people safe.

HOW IT WORKS

The planner uploads a TGS. An AI agent reads it, maps every barrier, sign, VMS board and detour, and tells the planner where on the street to look.

They scan those spots. Photogrammetry turns the photos into a 3D model of the real street, grounded by GPS.

An AI agent measures it against the plan and flags where the drawing gets the street wrong.

Simulation modelling predicts the knock-on effects: hours of delay, queue lengths and which streets take the detour. It's built on 2,700 real Melbourne closures and traffic signal counts, with a street-level simulation in testing. City of Melbourne pedestrian sensors add how many people walk through the site.

No survey crew, no consultants, no wait. Any planner with a phone can run it on any street.

WHAT'S BUILT SO FAR

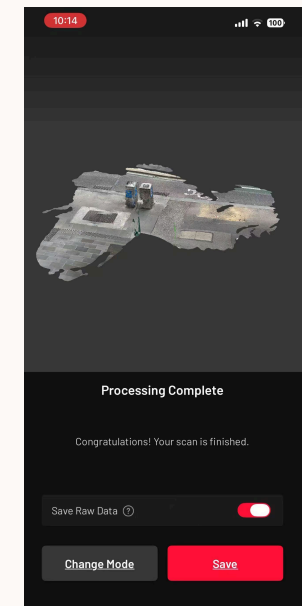
- Functional prototype at <https://taj-tea-m.vercel.app> with tech stack as Next.js, hosted on Vercel
- Explored and used SCATS signal counts, DTP signal sheets, OpenStreetMap for data
- AI agent and pipeline for TGS analysis (Agent running Claude Opus 5.5 through Vercel AI Gateway)
- Verified that performing photogrammetry using Scaniverse and exporting a LAS file works
- Verified that an AI agent is able to extract results of the scan & interpret the objects and key features
- Built impact models that are swappable, and every one returns the same result format.
 - A lookup built from about 2,700 past Melbourne closures (RADAR records matched to SCATS signal counts), plus an hour-by-hour queue calculation.
 - A SUMO traffic simulation fitted to measured counts at 9 junctions.

WHAT'S LEFT

- Stitching together individually functioning pieces of the pipeline into an end to end flow within Vercel
- Making it possible for users to have the model perform inference within the app on the cloud rather than in our local environments
- Conversion of the registration process of point clouds into a script



Scan in progress in app



Finished photogrammetry scan

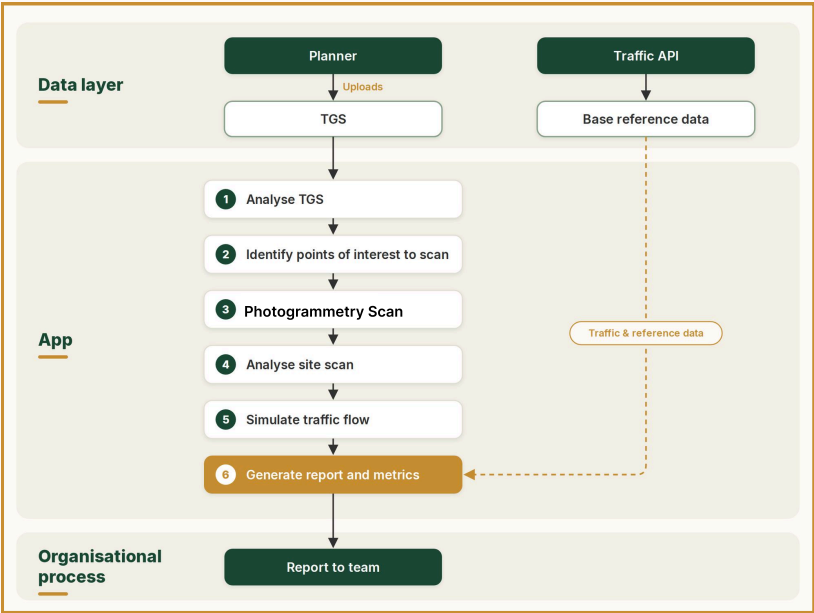
Visuals / UI / Diagrams

LOGO



DIAGRAM

Use-Case Flow



ICON



PALETTE



FRONT-END INTERFACE

Upload TGS Files & Scans



Site Impact Report

